Puspa Kamal Rai

9^{th} Mile, Thurbo, Mirik, Darjeeling - 734214, West Bengal | rpuspakamal@gmail.com | +91 90023 34495

Website | LinkedIn | GitHub

Vision

Driven by a passion for research and development, poised to make impactful contributions in future endeavors. Passionate and ambitious individual seeking to leverage my expertise in physics, music, coding, and leadership to contribute meaningfully to innovative projects and impactful organizations.

Career Objective

I have a strong interest in semiconductors and next-generation methodologies, but I am equally open to exploring any challenging opportunities that come my way. With skills in circuit design, embedded systems, semiconductor manufacturing, coding, managing datasets, and automation, I enjoy solving problems and take pride in working independently, thriving in dynamic environments where I can put my technical expertise to use. Committed to continuous learning and growth, I am looking for an internship or research program where I can make meaningful contributions while further developing my abilities.

Education

Sri Sathya Sai Institute of Higher Learning, M.Sc. in Physics — 2025 - 2027(expected)

- semester 1 ongoing

Sri Sathya Sai Institute of Higher Learning, BSc Hons in Physics — 2022 – 2025

- CGPA: 8.0
- **Coursework:** Classical Mechanics, Electromagnetism, Optics, Thermal Physics, Solid State Physics, Nuclear Physics, Quantum Mechanics, Atomic and Molecular Spectroscopy, Electronics-I and II, Microprocessors, Python Programming-I, Computational Techniques in Physics, Laboratory courses in Electronics, Mechanics, Optics, Electromagnetism, General Physics, Microprocessors, and Python Programming-II

Snowdrops School, CBSE 10th and 12th — 2020 and 2022

- 10th: 90.4%, 12th: 88%

Internships and Projects

UPS Designing, Self-Taken Project — Oct 2024 – Nov 2024

- Designed and built an innovative Uninterruptible Power Supply (UPS) capable of delivering stable and efficient AC output with a waveform resembling a smooth sine wave.
- Integrated the P55N06 MOSFET for high efficiency and low power loss, optimizing the system's performance under varying loads.

Automatic Bell System, Self-Taken Project — Mar 2025 – July 2025

- Developed an automated wireless alarming bell system to ensure seamless adherence to schedules and emergency needs.
- Integrated ESP01 Wi-Fi module and relay module, and built an interactive web interface using Svelte Kit for schedule management via server-side actions.

Water Purity Detection and Microbial Fuel Cells, Open-Ended Project — Feb 2025 – June 2025

- Improved water purity detection efficiency using DREAM Assay by building a low-cost spectrophotometer with Arduino.
- Worked on Microbial Fuel Cells, gaining insights into the intersection of physics and biology, applying critical thinking and problem-solving.

IIT Kharagpur, Internship — May 2025 – July 2025

- Under the guidance of Dr. Shiva Kiran Bhakta, developed a compact, low-cost, and efficient spectrometer using 3D-printed components and a smartphone camera as a sensor.

- Designed the system to incorporate photonic crystals and Bloch surface waves (BSW) for advanced spectral analysis.
- Implemented a modular adapter system to allow flexible analysis of reference and sample spectra, providing an accessible platform for optical diagnostics and educational tools.

Technologies

Languages: Python, Svelte Kit, HTML, CSS, JavaScript, SciLab, LaTeX

Technologies: Svelte Kit, Arduino, Raspberry Pi, ESP8266 , ESP32, LabView, LT-Spice, SQL, Power BI, MS based tools

Achievements and Extracurriculars

Flutist and Piccolo Player, University Brass Band — 2022 – Present

- Perform in the University Brass Band, blending precision and passion to deliver captivating melodies.

Vice Captain, Shanti House, Sri Sathya Sai Institute of Higher Learning — 2024-2025

- Led Shanti House to victory in the annual championship across all competitions.

2nd Place, Amaravati Quantum Hackathon, JNTUA Anantapuramu — September 23, 2025

- Achieved 2nd place in the semi-finals by implementing the BB84 quantum key distribution protocol using LabVIEW.